

Jai Malegaonkar
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EDUCATION

University of California, San Diego

Sep 2024 - Jun 2026

- MS in Computer Science and Engineering
- MicroMBA Certification

University of California, Davis

Sep 2020 - Jun 2023

- BS in Computer Science and BS in Computational Cognitive Neuroscience

SKILLS & INTERESTS

Programming: Python, C++, C, SQL

AI & Reinforcement Learning: RL, Multi-Agent RL (MARL), Gymnasium, Stable-Baselines3, JAX, PyTorch, TensorFlow, Intrinsic Motivation, Memory RL, Vision-Language-Action (VLA) Models

Robotics & Sim2Real: ROS/ROS 2, Nav2, GAMMS, OpenStreetMap (OSM), Gazebo, LiDAR & GNSS Fusion, Robot Localization, OpenCV

Machine Learning Engineering: Hugging Face, Scikit-learn, YOLO, Segment Anything Model (SAM), Contrastive Learning,

Cloud & Infrastructure: Docker, CI/CD, Linux (Ubuntu), Azure (Vision, DB, Repos), Git, Databricks, Spark, CUDA

Interests: Embodied AI, Social Curiosity, Reinforcement Learning, Social AI, Robotics

WORK EXPERIENCE

Cognitive Robotics Laboratory

Sep 2024 - Present

Graduate Researcher

- **Graph Based MultiAgent Simulator:** Developed a scalable graph based reinforcement learning simulator that can simulate games/scenarios on real world large graphs for 1000+ agents [GAMMS DOCS](#)
- **XOR Game Inspired MultiAgent RL:** Developing a randomized attention mechanism for MARL that dynamically breaks symmetries and coordination ties through stochastic masking, improving scalability and performance in environments with variable agent count
- **Sim2Real Using GAMMS:** Engineered a scalable Sim2Real research pipeline that decouples high-level strategic validation from physical deployment by prototyping Reinforcement Learning (RL) and heuristic strategies in low-fidelity graph simulations (GAMMS), utilizing OpenStreetMap (OSM) to automatically generate simulation topologies that map directly to real-world coordinates. This architecture enables a seamless transition from "in-silico" strategy to "in-field" execution by translating abstract graph-based decisions into actionable ROS 2 Nav2 goals via a GPS-integrated deployment bridge, allowing researchers to validate and deploy complex multi-agent behaviors with zero changes to the core logic.

Leidos

Apr 2025 - Sep 2025

AI/ML Research Intern

- **Engineered** a physics-informed data pipeline to process raw TLEs, using an SGP4 propagator and orbital mechanics principles to extract a rich, n-dimensional feature vector that grounded the model in physical reality and enabled robust training.

- **Architected** a novel two-stage deep learning pipeline to discover satellite networks, first learning satellite identity embeddings via Supervised Contrastive Learning and then predicting links with a hybrid physics-embedding classifier
- **Developed** a temporal analysis pipeline using Hidden Markov Models (HMMs) to filter noisy predictions, successfully identifying coherent "episodes" of coordination and enabling the discovery of long-term operational campaigns from raw data.

Volvo Group

Aug 2023 - Aug 2024

AI/ML Engineer Intern

- **Flow Lane Detection:** Built real-time computer vision systems using Azure Vision and YOLO, applying transfer learning and hyperparameter tuning techniques to improve accuracy, find anomalies, and reduce latency.
- **Chassis Lock Detection:** Finetuned a pre-trained SAM model by retraining the final layers, leading to improved chassis lock detection accuracy in production.
- **Subcontractor Price Analysis:** Developed a library and performed analysis on price calculations from third-party vendors, confirmed numbers by doing calculations, and corrected and revised discrepancies, optimizing and reducing expenses using C#.

Python Engineer Intern

- Developed a data pipeline for collecting and processing truck signal data, leveraging models such as Random Forests and LSTMs to classify signal anomalies and predict driver behavior, improving maintenance scheduling and reducing vehicle downtime by 15%.

TEACHING EXPERIENCE

UCSD PSYCH Department - Teaching Assistant

Sep 2024 - Dec 2024

Mathnasium - K-12 Math Instructor

Sep 2020 - Jun 2023

PUBLICATIONS

- Rohan Patil, **Jai Malegaonkar**, Xiao Jiang, Andre Dion, Gaurav S. Sukhatme, Henrik I Christensen, GAMMS: Graph based Adversarial Multiagent Modeling Simulator. (Accepted / NeuS), 2026.
- Rohan Patil, **Jai Malegaonkar**, Henrik I Christensen, Randomness is sometimes necessary for coordination. (Under Review /NeurIPS), 2026.

CURRENT PROJECTS

- **H-SW-VIME:** Developing a novel intrinsic motivation framework using a Wyner Bottleneck to extract common information between episodic history and state transitions, filtering out environmental noise (the "Noisy TV" problem). Implemented a Spherical Cauchy VAE with a differentiable reparameterization trick to map transition dynamics onto hyperspherical manifolds, enabling robust exploration and discovery of complex causal chains in sparse-reward environments.
- **Learning Transferable Robot Guidance Behaviors from Human Video Demonstration:** Designed a controlled human-demonstration protocol and engineered an end-to-end video extraction pipeline—leveraging pose estimation, spatial tracking, and visual scene understanding—to translate natural human guidance behaviors into actionable semantic inputs for Vision-Language-Action (VLA) models.